

Notation as a Tool of Thought - Adjunction

Adjunction: $F(x) \leq y$ iff $x \leq G(y)$, given F, G monotonic.

$$\begin{aligned} F(x) \leq F(x) &\implies x \leq G(F(x)) \implies F(x) \leq F(G(F(x))) \\ G(y) \leq G(y) &\implies F(G(y)) \leq y \implies G(F(G(y))) \leq G(y) \end{aligned}$$

1 there are both function application and composition above, we can get rid of application by replacing x with constant function $X : * \rightarrow x$. $F(x) = F \cdot X$

2 use diagram order $F \cdot X \implies XF$

3 replace F with $/$, G with $\backslash$

4 put $a \leq b$ as a over b. $\boxed{\frac{a}{b}}$

5 change from $X \leq XF$ to $1 \leq F$

rewrite

$$\boxed{\frac{X/}{Y}} = \boxed{\frac{X}{Y\backslash}}$$

adj

$$\boxed{\frac{\backslash X}{Y}} = \boxed{\frac{X}{/Y}}$$

ladj

$$\begin{aligned} \boxed{\frac{X/}{X/}} &\xRightarrow{\{\text{adj}\}} \boxed{\frac{X}{X\backslash}} \xRightarrow{\{\text{rewrite}\}} \boxed{\frac{*}{\bigwedge}} \xRightarrow[\text{unit}]{} \begin{aligned} &\xRightarrow{\{\text{put } \backslash \text{ on left}\}} \boxed{\frac{\backslash}{\bigwedge}} \\ &\xRightarrow{\{\text{put } / \text{ on right}\}} \boxed{\frac{/}{\bigwedge}} \end{aligned} \\ \boxed{\frac{Y\backslash}{Y\backslash}} &\xRightarrow{\{\text{adj}\}} \boxed{\frac{Y\bigvee}{Y}} \xRightarrow{\{\text{rewrite}\}} \boxed{\frac{\bigvee}{*}} \xRightarrow[\text{counit}]{} \begin{aligned} &\xRightarrow{\{\text{put } / \text{ on left}\}} \boxed{\frac{\bigvee}{/}} \\ &\xRightarrow{\{\text{put } \backslash \text{ on right}\}} \boxed{\frac{\bigvee}{\backslash}} \end{aligned} \end{aligned}$$

now we have $\bigvee = /$ and $\bigwedge = \backslash$, and you just discovered string diagrams.

§1. Division

$RS \leq T$ iff $R \leq T/S$ is the same law, with $/ = \bullet S$ and $\backslash = \bullet /S$.

§1.1. $R/(S_1S_2) = (R/S_2)/S_1$

Take two, $/ = S_1$ and $/ = S_2$. Each one climbs a level — T , TS_1 and TS_1S_2 are three different hom-sets — so the composite is one stick twice as long, and adj walks the two down one at a time:

$$\begin{array}{ccccccc}
 \boxed{\begin{array}{c} T \\ \backslash \\ R \end{array}} & \xlongequal{\{\text{adj}\}} & \boxed{\begin{array}{c} T \\ / \\ R \end{array}} & \xlongequal{\{\text{adj}\}} & \boxed{\begin{array}{c} T \\ / \\ R \end{array}} & \xlongequal{\{\text{adj}\}} & \boxed{\begin{array}{c} T \\ / \\ R \end{array}} \\
 T \leq R/(S_1S_2) & & TS_1S_2 \leq R & & TS_1 \leq R/S_2 & & T \leq (R/S_2)/S_1
 \end{array}$$

The left box is also $T \leq R/(S_1S_2)$: one arrow or two, the wire is the same. So both right-hand sides hold of exactly the same T , and $R/(S_1S_2) = (R/S_2)/S_1$.

§1.2. $f(R/S) = (fR)/S$

ladj with $/ = f$ and $\backslash = f^\circ$ — a map is a left adjoint, so it moves at the left end. Crossing the bar turns $/$ into $\backslash$ either way; at the right end the stroke travels down, at the left end up, and it never changes ends. The other reading, $fX \leq Y$ iff $X \leq f^\circ Y$, would need $1 \leq f^\circ f$ — f surjective.

$$\begin{array}{ccccccc}
 \boxed{\begin{array}{c} T \\ / \\ R \end{array}} & \xlongequal{\{\text{ladj}\}} & \boxed{\begin{array}{c} T \\ \backslash \\ R \end{array}} & \xlongequal{\{\text{adj}\}} & \boxed{\begin{array}{c} T \\ \backslash \\ R \end{array}} & \xlongequal{\{\text{ladj}\}} & \boxed{\begin{array}{c} T \\ / \\ R \end{array}} \xlongequal{\{\text{adj}\}} \boxed{\begin{array}{c} T \\ / \\ R \end{array}} \\
 T \leq f(R/S) & & f^\circ T \leq R/S & & f^\circ TS \leq R & & TS \leq fR \quad T \leq (fR)/S
 \end{array}$$

The two ends are one picture read with the brackets in the two places they can go, and they hold of the same T . Hence $f(R/S) = (fR)/S$.